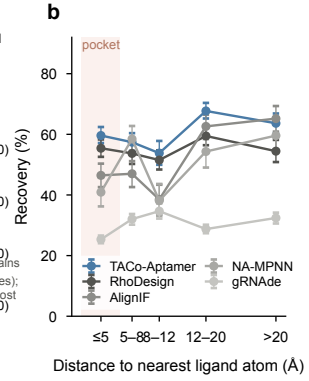
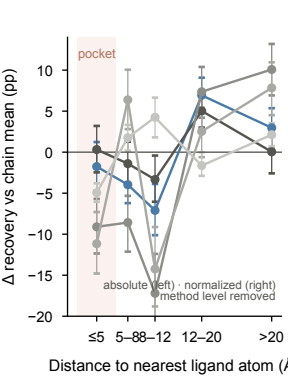
a

Model variar	Global best (%)	Interface best (%)	Δ (iface−Iface overall, pp)	Δ vs full (pp, P)
Full model (ens	74.58	85.60	+10.05	—
– ensemble (sin	71.42	81.35	+9.08	-4.25 (P=0.0000)
– SS condition	70.59	81.12	+9.80	-4.48 (P=0.0000)
– data augmenta	70.66	79.29	+7.93	-6.31 (P=0.0000)
– ligand conte	70.78	80.92	+8.71	-4.68 (P=0.0000)

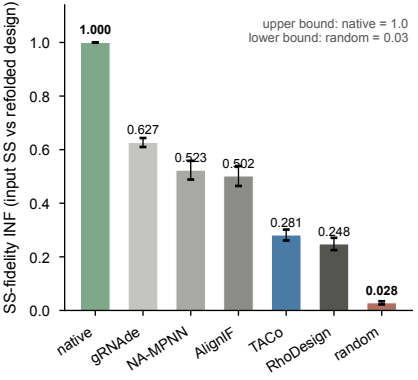
300-sample pools (3 T × 100) · final noiface recipe · global = test 477 chains
interface = 115 ligand-contact chains (<5 Å; 100 with ≥3 interface residues);
rows 2–5 are single models — within that family, component removals cost
2.5–6.3 pp globally (interface effects n.s. at this test size)

b

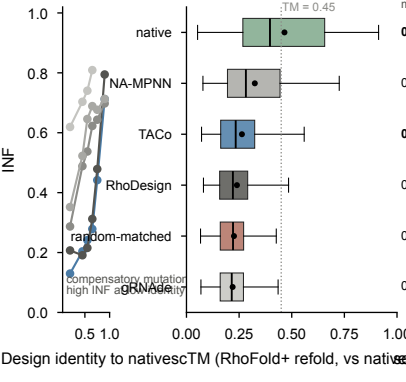




c



f



g

